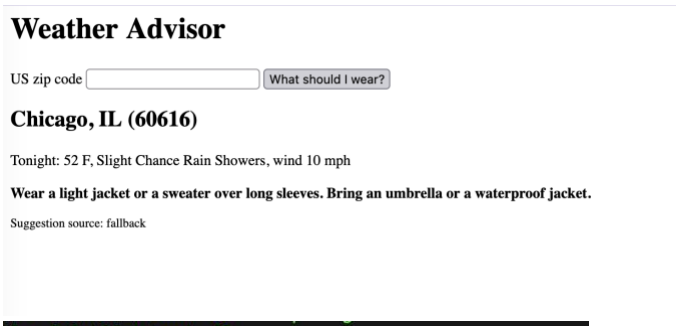
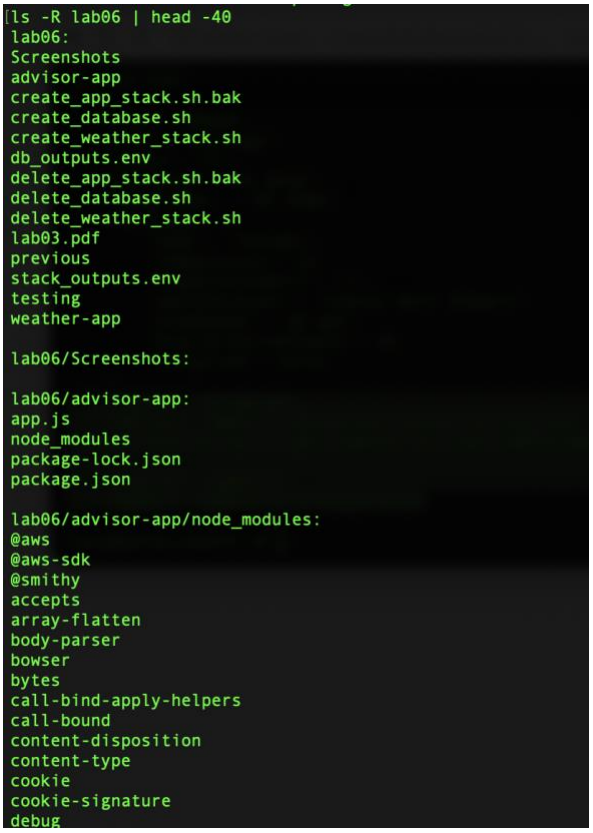
Lab 06 Reflection Screenshots and explanations

Weather Advisor

US zip code What should I wear?





Lab 06 Reflection Screenshots and explanations

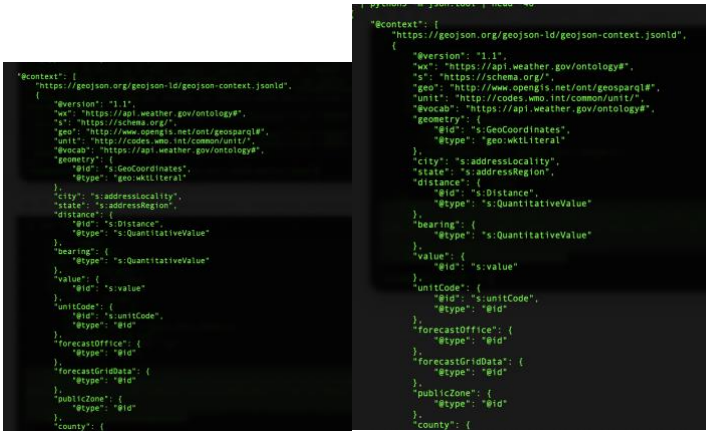

```
{
    "country": "United States",
    "country abbreviation": "US",
    "post code": "60616",
    "places": [
        {
            "place name": "Chicago",
            "longitude": "-87.6306",
            "latitude": "41.8426",
            "state": "Illinois",
            "state abbreviation": "IL"
        }
    ]
}
```


```
root@8d53bccd4076:/cloud-computing-lab-test# curl -s -o /dev/null -w "%{http_code}\n" https://api.zippopotam.us/us/00000
404
```

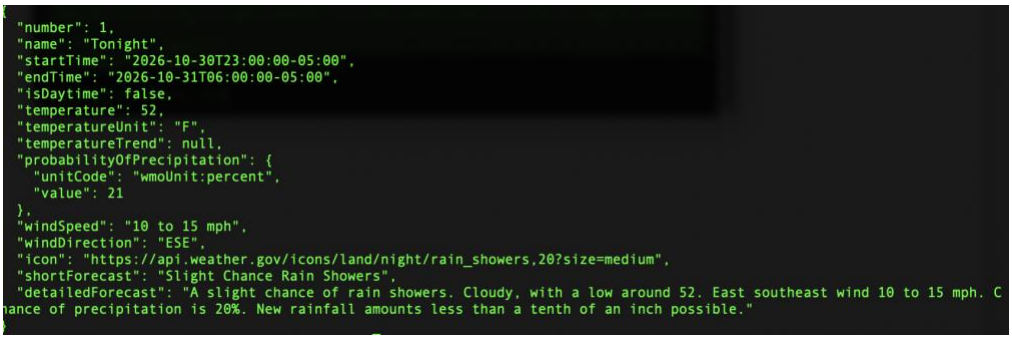

```
{
    "zip": "60616",
    "city": "Chicago",
    "state": "IL",
    "latitude": "41.8426",
    "longitude": "-87.6306",
    "forecast": {
        "name": "Tonight",
        "temperature": 52,
        "temperatureUnit": "F",
        "shortForecast": "Slight Chance Rain Showers",
        "windSpeed": "10 to 15 mph",
        "precipitationChance": 22,
        "isDaytime": false
    },
    "cache": "miss",
    "advice": "Wear a light jacket to stay warm in the 52 degree weather. You should also bring an umbrella just in case
of light rain showers.",
    "source": "gemini",
    "model": "gemini-3.1-flash-lite"
}
```


```
PASS weather tier returns a forecast status 200
PASS a repeat call is answered from the database (cache hit) cache=hit
PASS history lists the saved lookups status 200, count=6
PASS weather tier rejects a bad zip with 400 status 400
PASS weather tier returns 404 for an unknown zip status 404
PASS advisor tier returns clothing advice status 200, 8.4s
  source=gemini model=gemini-3.1-flash-lite
  advice: Wear a light jacket to stay comfortable in the cool evening air. You may want to carry a small umbrella just in
 case of light rain.
PASS advice came from Gemini (not the fallback) fallback means the key, quota or model name needs a look
PASS advisor tier rejects a bad zip with 400 status 400
PASS advisor home page loads status 200

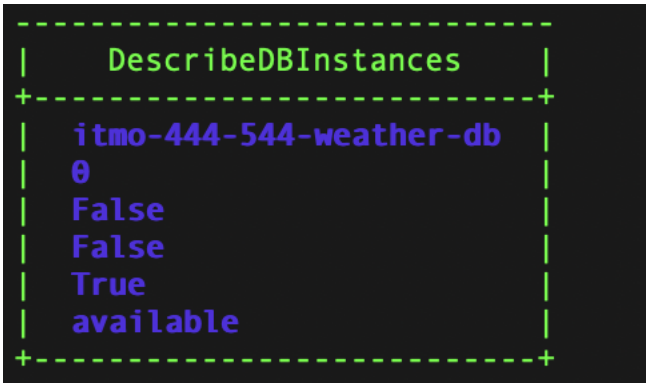
9 of 9 checks passed
```



Lab 06 Reflection Screenshots and explanations

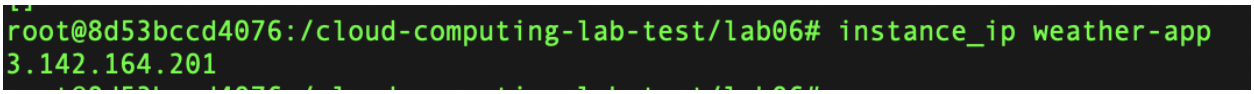


The weather service requires two API calls because it it uses latitude and longitude rather than zip codes to retrieve weather data. The first API call obtains the geographic coordinates, and the second uses those coordinates to retrieve the weather forecast.

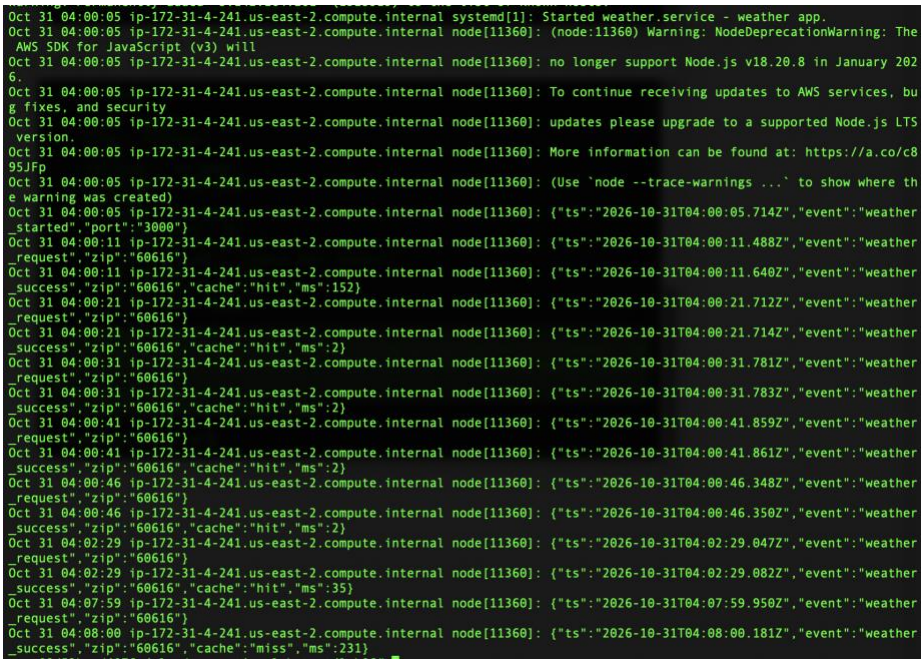




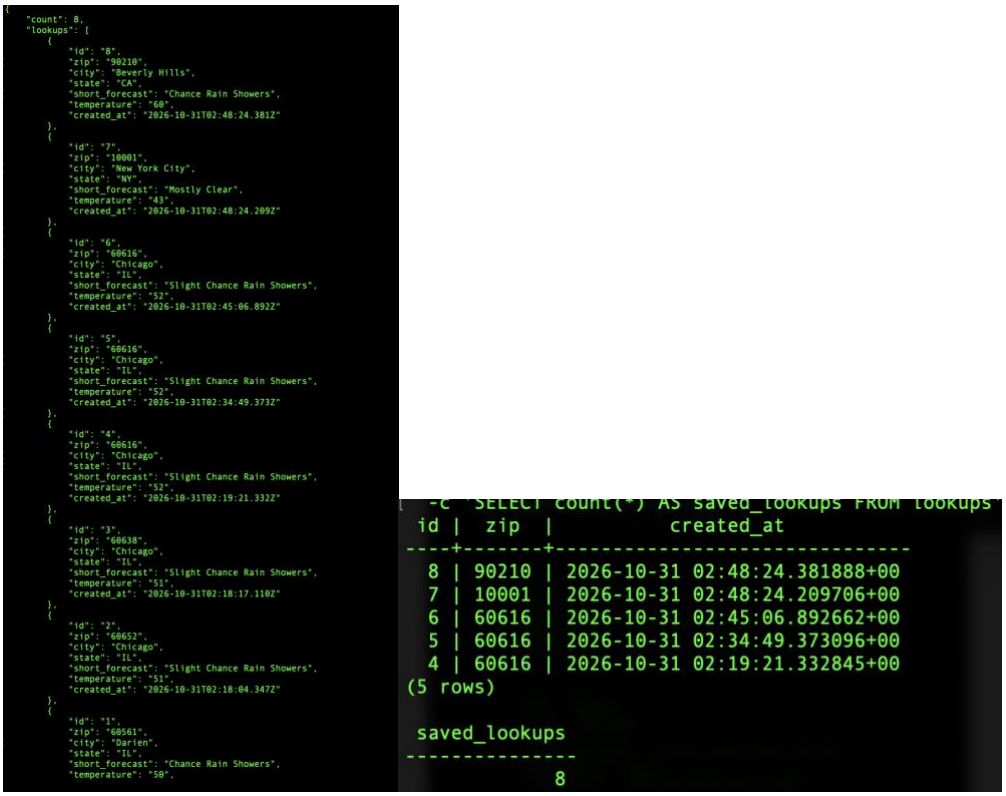




Lab 06 Reflection Screenshots and explanations

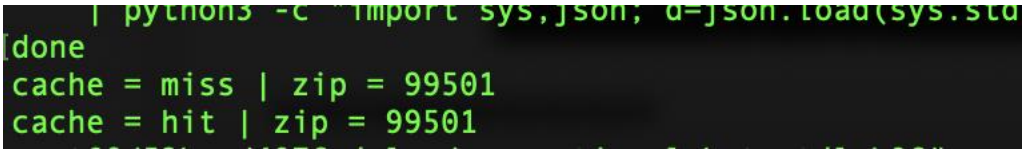


The logs confirm that the Weather Service successfully processed the request for the zipcode 60616 and retrieved the data from the cache.

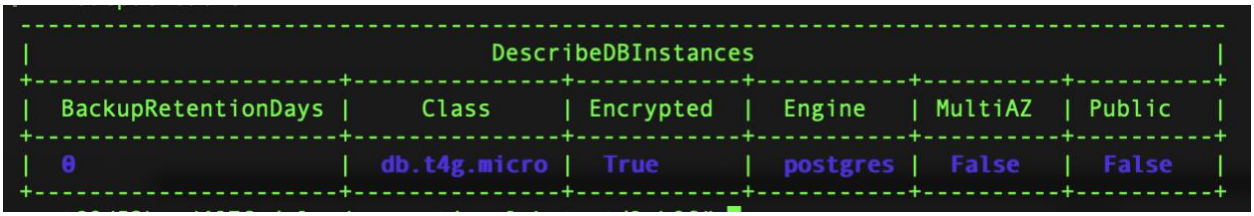


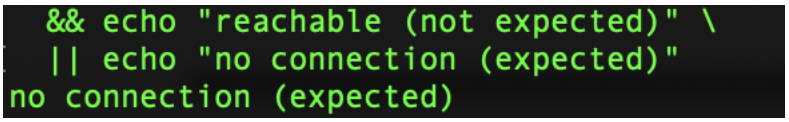
The weather service uses caching to store previously retrieved weather data, which explains why there are fewer database records than incoming requests. When the same

information is requested again, the service retrieves the cached data instead of making another API call.



Caching reduces the number of requests to external APIs by reusing previously retrieved data, since APIs limit how many requests you can make. This is beneficial because many APIs have rate limits that restrict how frequently the requests can be made.







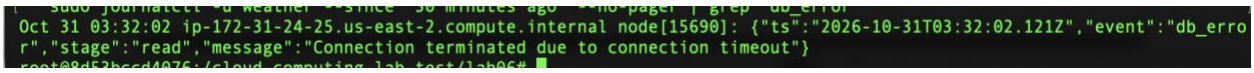
If the Advisor tier was compromised by an attacker, they could have access to the Gemini API key because the Advisor's IAM role has the permission to read it. The role cannot access the PostgreSQL database password via the Systems Manager because it does not have permission that allow access to that secret. This separation follows the principle of least privilege and helps limit the impact of a security breach.



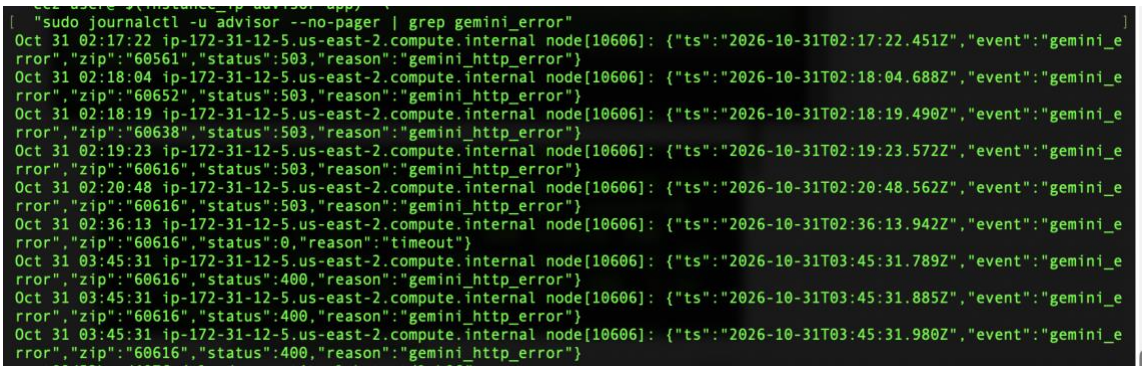


Using security group references instead of specific IP addresses allows the application instances to continue receiving traffic from the load balancer, when if the IP addresses change. This provides greater flexibility allows for better communication among the load balancer and the application instances.

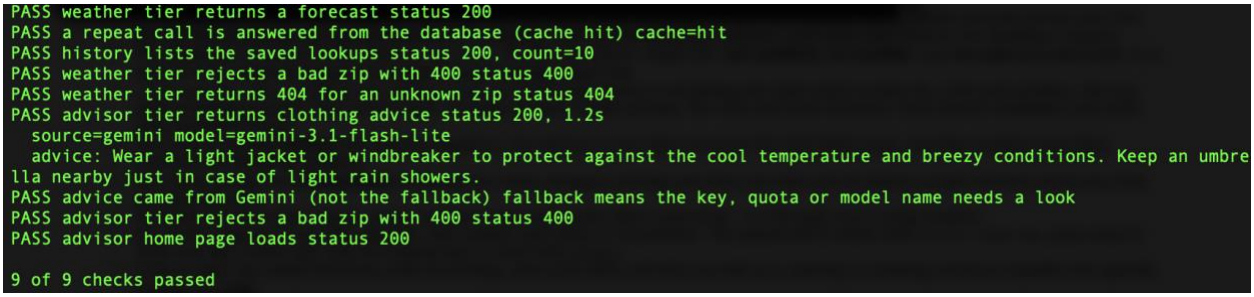
Lab 06 Reflection Screenshots and explanations

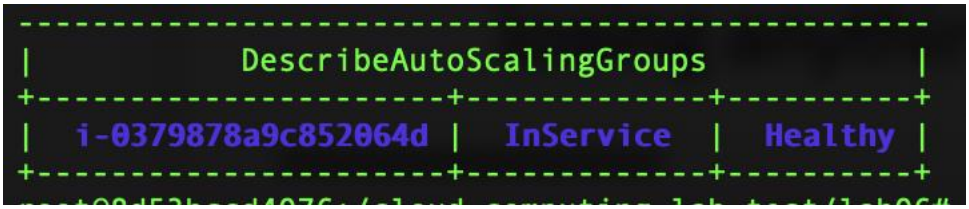


A 30 second pause prevents the Weather Service from continuously trying to connect to the PostgreSQL while the database is unavailable. Without a pause the requests could trigger a failed connection attempt, causing delays.



Users may not realize that Gemini is experiencing issues because the application continues to function and provides users advice because of the fallback mechanism. If the logs are not regularly monitored, the underlying problem could go unnoticed and remain unresolved.







By saving data in the database instead of the EC2 instance new instances can be terminated and replaced without causing disruptions. Because the weather history is saved in the PostgreSQL database the new instances have access to the same data without losing anything.

Lab 06 Reflection Screenshots and explanations



When I terminated the Weather EC2 instance the advisor returned an HTTP 502 error with the message weather_service_unavailable. Auto scaling replaced the instance and it was recovered after 122 seconds. The history was not lost because the information was stored on the PostgreSQL database and not the EC2 instance.

